Complex Tables and Fat Cats

FCAT 1190 Problem Set 1

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Not All Cats

According to Dr. McDaniel, VMD and the Cornell Feline Health Center, roughly 5-10% and 40% of the cat population can be considered overweight and obese, respectively. Fatter cats are at higher risk of multiple chronic diseases, such as:

- `Heart Disease`
- `Asthma`
- `Arthritis`
- `Diabetes`
- `Chronic Dermatitis`

all of which are incurable.

Early detection of the overweight condition in cats is therefore necessary to prevent a chronic condition. Detection relies on qualitative measures such as the body condition score, which grades the silhouette of a cat against those of archetypal body shapes and may include other measures such as palpation of the ribcage to examine the thickness of the tissue there.

The point of this assignment is to understand the efficacy of these qualitative measures under commonly confounding factors of body length and your cat being a baby.

*Who is neither a statistician nor a biologist, but loves accessibility and cats.

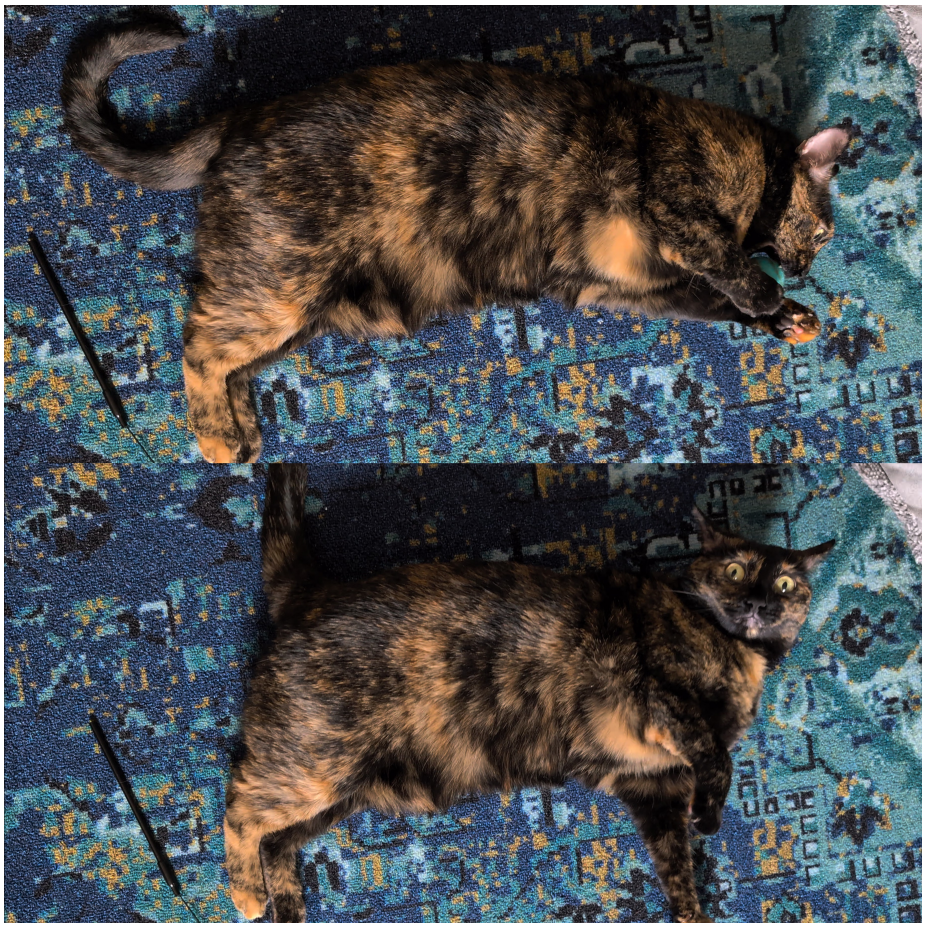


Figure 1: Smidge is a cat who is long, a baby, and possibly fat. The latter is obviously not my fault, as being a long baby has made it statistically unlikely that I will be able to determine if she is fat or not[1].

Motivated by the formal case study in fig. 1, we present the following measurements

[1] Smidge has since been on a diet and is now healthy.

Table 1: Table 1: Measures probabilities of achieving a correct body condition score among independent samples

Categories		Body Length		Totals
		Short	Long	
Baby?	Pays Rent	0.62	0.11	0.73
	A Baby	0.1	0.12	0.15
	Gremlin!!	0.28	0.16	0.44

Question 1

What are potential sources of 'probability leak' in this data?

Question 2

Using the data in the table, what is the likelihood that being long and being a gremlin are not independent?

Question 3

Treating the values in tbl. 1 as means of a very broad sample, derive probability that a cat owner will not be able to tell whether their cat belongs to *any* of the categories in tbl. 1.